Nocturia & Male Incontinence Surgery Rapid Revision

Campbell-Walsh-Wein · Chapters 112, 123

For MCh Urology theory & viva · NEET-SS / competitive MCQs

How to use (total ≈ 1h 17min even at a slow, careful pace)

- Each chapter reads top-to-bottom in the order you would write an essay answer: definition → pathology → diagnosis → management → complications → follow-up. The same material serves theory and viva.
- Boxed **ALGORITHM** and shaded **KEY POINT** panels carry the reasoning examiners ask about ("why do we…?").
- **MCQ one-liners** end each chapter: re-read these the day before the paper.
- The last section puts the big treatment choices side by side for essay questions.

Conventions: **underlined bold** = number to memorise · MCQ = exam trivia only · → *Ch 123* = cross-reference · † = not in the source chapter notes (added because it is essential in practice). Abbreviations are spelled out at first use in each chapter.

Chapter	Topic	Slow-reader time
112	Nocturia	48 min
123	Surgical Procedures for Sphincteric Incontinence in the Male	26 min
★	Putting it together: nocturia and male stress incontinence	3 min
	Total (5327 words, at 70 words/min)	**1h 17min**

112 Nocturia

Reading time ≈ 48 min (slow pace)

1 · DEFINITION AND TERMINOLOGY

- **Nocturia** — International Continence Society (ICS) 2019: the number of times urine is passed during the **main sleep period**, from the time a person intends to sleep until the time they intend to rise. The period is set by the **individual's own schedule** (shift workers included), not the clock.
- It does **not exclude 1 void per night** and does **not incorporate bother**.
- ICS 2002 said "wake at night **to void**", implying urge-driven waking differs from poor sleep. ICS 2019 dropped the cause: urge and poor sleep are intertwined, and sleep or "non-urological" causes are **leverage for interdisciplinary care, not grounds to exclude** a patient.
- Guidance comes from the European Association of Urology (EAU; standalone 2017 guideline and nocturia sections in Male and Female lower urinary tract symptoms [LUTS] guidelines) and ICS. The American Urological Association (AUA) has none; its overactive bladder (OAB) guideline calls **isolated nocturia a separate entity**.

Key diary terms

Term	Definition / formula	Meaning
Actual number of nightly voids (ANV)	Voids within the main sleep period	Severity count
First uninterrupted sleep period (FUSP)	Sleep onset → first night void (h)	Key sleep measure
Nocturnal urine volume (NUV)	All night voids **+ first morning void**	Volume made during sleep
Nocturnal polyuria index (NPi)	NUV / total 24-h volume (TUV)	Proportion made at night
Nocturnal urine production (NUP)	NUV / hours of sleep period	Hourly rate
Maximum voided volume (MVV); nocturnal MVV (NMVV)	Largest void in 24 h; in sleep period	Proxies for functional capacity
Nocturia index (Ni)	NUV / MVV	>1: night output exceeds capacity
Modified Ni (MNi)	NUV / NMVV	Night data only
Predicted nightly voids (PNV)	Ni − 1	—
Nocturnal bladder capacity index (NBCi)	ANV − PNV	>0: voids below capacity

Trap: the first morning void is **split** — its volume goes into NUV, but it counts as daytime frequency, not a nocturia episode.

2 · WHY NOCTURIA MATTERS

- **Common**: up to **3 in 5** people >70 years void ≥2 times nightly; age is the most robust correlate. Boston Area Community Health survey MCQ: higher in Black and Hispanic people, partly via socio-economic status.
- **Bothersome**: most with ≥**2** voids are bothered, but one void can be very bothersome — **any nocturia that brings a patient to a urologist is clinically significant**. It may dominate the interview and mask other LUTS.
- **Sleep**: urge-driven waking is the **commonest cause of adult sleep disruption**. Slow-wave sleep (N3), the deepest and most restorative stage, clusters in the early night and is not recovered later; most first voids occur within **2–3 h** of sleep onset, so they land in it. Non-rapid eye movement sleep is ≈75% of sleep, in 4–6 cycles nightly, but slow-wave sleep is only 10–25%. Hence daytime fatigue

even when total sleep time is preserved, and a shorter time to first void means worse whole-night sleep.

- **Falls and hip fracture**: nocturia is among the LUTS most strongly linked; older adults are most affected and least steady in the dark.
- **Systemic disease and mortality**: nocturia may herald cardiometabolic (strongest evidence), renal, endocrine or neurological disease. ≥2 voids per night is associated with higher mortality; causality unproven — probably via falls, poor sleep and comorbidity.

> **KEY POINT** Nocturia matters because of when it strikes and what it signals: the first void breaks restorative slow-wave sleep, and the symptom may be the visible edge of systemic disease.

3 · PATHOPHYSIOLOGY — PRODUCTION, CAPACITY AND SLEEP

Aetiology is multifactorial — **"Sleep CALM"**: **S**leep disorders, **C**omorbidities, **A**ctions (behaviour, lifestyle), **L**ower urinary tract dysfunction (LUTD), **M**edications. Idiopathic nocturia is small but real.

- **Common mechanism**: every factor upsets the balance of **night urine production**, **functional capacity** and **sleep architecture**.
- Production is continuous; after emptying at bedtime, volume climbs to capacity, and then there are only two endpoints — **wake to void or enuresis**. Sleeping through requires night production ≤ capacity.
- **Sleep, the third arm**: sleep disorders cause "convenience" voids while already awake, may raise production and lower the arousal set-point. The relation is **bidirectional**.

Nocturnal polyuria (NP) and global polyuria

Criterion	Threshold	Comment
NPi (ICS)	>**20%** young to >**33%** older adults	Most used; poor discrimination
NUP (ICS alternative)	>**90 mL/h**	More specific for production-driven nocturia
Global polyuria	>**40 mL/kg/24 h** (>2.8 L at 70 kg)	Uncommon, but serious differential — **always investigate**

- **Krimpen study** MCQ: NPi >33% in 92% of older men with ≥2 voids but also 70% of men **without** nocturia.
- Excess night production contributes to **most adult nocturia**. NP, global polyuria and small capacity are not mutually exclusive and often coexist.

Functional capacity and the mismatch

- True capacity = voided volume + post-void residual (PVR); MVV/NMVV are proxies. No consensus: normal usually >300–400 mL; **150–300 mL** suggests diminished capacity — either **24-h** (all voids small) or **night-specific** (small night voids, normal day voids; partly convenience voids).
- Waking depends on the **ratio** of production to capacity: **Ni is the strongest diary determinant of severity and treatment response**. PNV = Ni − 1 because the last night volume leaves as the first morning void.

ALGORITHM
NUV 800 mL, MVV 200 mL → Ni = 4 → PNV = 3 → diary shows 6 night voids → NBCi = **3** (voiding well below capacity)

Investigate diminished night capacity when NBCi >**1.3**; NBCi >**2** is strongly associated with severe nocturia.

4 · THE BLADDER DIARY

The guideline cornerstone, objectively showing the production–capacity–sleep relationship. EAU: recommended for female LUTS and for male LUTS with prominent storage symptoms or nocturia (strong).

ALGORITHM
Measuring container → time and volume of every void → time, volume and type of every drink → mark bedtime (intending sleep) and rising → urgency grade and leakage per void

- **Frequency–volume chart** = void times and volumes only; **bladder diary** adds intake, pads and incontinence.
- International Consultation on Incontinence diary urgency codes: **0** no need (social void), **1** normal desire, **2** urgency that passed, **3** urgency, dry, **4** urgency with leak.
- **Diary ANV is the gold standard for severity**; patients usually overestimate on questionnaires. **Intent governs**: lying awake intending to sleep is still the sleep period.
- **FUSP is the most important sleep measure**: inversely related to night voids, and its prolongation marks successful treatment with better sleep and daytime function — an outcome measure and target.
- Duration: EAU **3 days** (Male: at least 3). Longer diaries are more reliable but adherence falls ("diary fatigue"); completed by fewer than half of urology patients.
- **Nocturnal-only diary** gives ANV, FUSP, NUV, NUP, NMVV and MNi (reasonable surrogate for Ni), but **cannot diagnose global polyuria** — the diagnosis most tied to serious disease. Partial work-around: hourly, the global polyuria threshold is 90 mL/h at 54 kg but 167 mL/h at 100 kg, so very high night volumes in heavy patients may signal 24-h polyuria. Unvalidated; not recommended for women by EAU; better only than no diary.

KEY POINT Every cause works through one equation: production against capacity, gated by sleep. The diary measures both sides, and their ratio (Ni) predicts severity better than either alone.

5 · EVALUATION — THE SLEEP CALM WORK-UP

History, examination and basic tests in all → tailored ancillary tests and multidisciplinary referral. A validated questionnaire helps at baseline and follow-up; **TANGO** (Targeting the individual's Aetiology of Nocturia to Guide Outcomes) screens all domains, including falls.

Factor	History / examination	Tests (basic → ancillary)
Sleep	Sleep habits, apnoea, restless legs, pain, anxiety, depression	STOP-BANG, sleep diary → polysomnography, sleep specialist
Comorbidities	Full history; blood pressure, volume overload, neurological signs	Metabolic panel → glycated haemoglobin (HbA1c), electrocardiogram (ECG), echocardiography, natriuretic peptide, albumin:creatinine ratio (ACR), calcium, thyroid, water deprivation test
Actions	Fluid, alcohol, caffeine, sodium; body mass index (BMI)	Fluid intake diary
LUTD	Concurrent LUTS, prior surgery; abdominal, genital, rectal examination	Urinalysis, PVR → prostate-specific antigen (PSA), uroflowmetry, cystoscopy
Medications	Reconcile drugs and **dose timing**	—

- **Trap**: the urological row is one of five; stopping there misses four-fifths of the differential.
- First-line management = **optimise every Sleep CALM abnormality found**.

6 · SLEEP DISORDERS

Obstructive sleep apnoea (OSA) is the best-described link: incidence rises with nocturia frequency, and frequent nocturia predicts severe OSA.

Arm	OSA mechanism	Continuous positive airway pressure (CPAP)
Production (main)	Hypoxia → pulmonary vasoconstriction → ↑ right atrial transmural pressure → ↑ **atrial natriuretic peptide (ANP)** → natriuresis	↓ night production
Capacity	Intermittent hypoxia → oxidative stress → detrusor instability	Less detrusor overactivity
Sleep	More light sleep, less slow-wave sleep	↑ slow-wave sleep

- CPAP reduces nocturia by up to ≈**2 voids** in moderate-to-severe OSA — among the **largest effects of any intervention**. EAU Female (strong): suspected OSA → sleep clinic.
- **STOP-BANG**: Snoring, Tired, Observed apnoea, Pressure (hypertension), BMI >**35**, Age >**50**, Neck >**40 cm**, Gender male; ≥**3** = high risk; highly sensitive for moderate-to-severe OSA.
- Establish the patient's **primary motive for getting up** (urge versus already awake). Ask about choking arousals, witnessed apnoeas and morning headache (OSA); an evening urge to move legs relieved by movement (restless legs); and odd night events (parasomnias).
- **Melatonin** may suit older adults, who are most vulnerable to hypnotics, anticholinergics and antidiuretics (limited evidence). Behavioural insomnia treatment improved sleep and reduced night voids. Involve sleep experts when indicated.

> **KEY POINT** OSA is the model for the whole chapter: one disorder raises production, lowers capacity and fragments sleep — which is why treating it gives one of the largest nocturia responses.

7 · COMORBIDITIES

- Medical disease acts mainly through **excess night production**. Nocturia may be the **first sign** of hypertension, metabolic syndrome, coronary disease, diabetic end-organ damage or early kidney disease. EAU: consider systemic causes in all; even **non-bothersome nocturia merits a frequency–volume chart**.
- **Volume overload**: fluid third-spaced into the legs by day is mobilised on lying down, adding to NUV — exaggerated in heart failure, where diuresis reduced nocturia with oedema.
- **Hypertension**: blood pressure normally dips **10–20%** in sleep; **non-dipping** probably mediates the link via renin–angiotensin–aldosterone dysregulation → **nocturnal natriuresis**. Hydrochlorothiazide improved nocturia in men with α-blocker-refractory LUTS.

Disorders of water homeostasis

Arginine vasopressin (AVP): supraoptic and paraventricular nuclei → posterior pituitary → released mainly by rising osmolality → **V2 receptors** on collecting-duct principal cells → water reabsorption. Defects show on the diary as **24-h polyuria**.

Condition	Defect	Causes
Central diabetes insipidus (DI)	↓ AVP secretion	Brain tumour, haemorrhage, trauma, autoimmune
Nephrogenic DI	Renal AVP insensitivity	Drugs, electrolyte disturbance, renal failure
Primary polydipsia	AVP suppressed by intake	Idiopathic; psychiatric

Arginine-stimulated **copeptin** (a cleavage product of the vasopressin precursor) appears superior to the water deprivation test. Every global polyuria needs evaluation to avoid missing a neoplastic, traumatic, inflammatory or psychiatric process; cause-specific treatment normalises output.

> **KEY POINT** Comorbid disease reaches the bladder mostly through the kidney — third-spaced fluid, nocturnal natriuresis and failed antidiuresis all raise night production. Global polyuria and non-bothersome nocturia are both reasons to look beyond urology.

8 · ACTIONS — LIFESTYLE AND BEHAVIOUR

- Trial evidence is thin and epidemiology inconsistent (patients self-restrict), but physiology is strong — **timing matters more than total amount**: a water bolus causes diuresis within ≈1 h; caffeine and alcohol within 2–3 h (caffeine half-life 2.5–4.5 h) and both disturb sleep; a protein-rich evening meal causes hyperfiltration for 2–3 h.
- **Practice**: restrict fluid, caffeine and alcohol from **3–4 h** before sleep; older adults should drink more by day to avoid dehydration. Patients with high baseline intake gain most.
- Also: **sodium** restriction; weight loss and exercise; **compression stockings** (less daytime leg fluid to mobilise); bladder training, pelvic floor muscle training (PFMT), treating constipation.

9 · LOWER URINARY TRACT DYSFUNCTION

- Nocturia accompanies bladder outlet obstruction (BOO), OAB (up to **85%** of trial participants), detrusor underactivity, neurogenic bladder and the genitourinary syndrome of menopause (GSM) → *Voiding Dysfunction book, Ch 110*.
- **Isolated nocturia is not a LUTS** and is not expected to respond to bladder or prostate drugs; use them when LUTD is genuinely suspected.
- Assessment: urinalysis in all (treat infection, repeat); **PVR in both sexes**; PSA only if it changes management. †Urodynamics is not routine, only when LUTD needs defining → *Voiding Dysfunction book, Ch 107*.

Benign prostatic obstruction

- Relieving BOO should help by lowering PVR and afferent input. Nocturia data are secondary endpoints of benign prostatic hyperplasia (BPH) trials: **α-blockers** carry most of the effect; 5α-reductase inhibitors (5ARIs) alone are weakest. Gains are fractions of a void with a large placebo response — **statistical, not clinical, significance**. Drugs and dosing → *Voiding Dysfunction book, Ch 113*.
- Phosphodiesterase type 5 inhibitors (PDE5Is): EAU — **do not offer** for nocturia.
- **Surgery**: transurethral resection of the prostate (TURP) reduced nocturia more than microwave therapy, α-blocker or watchful waiting, but sleep and quality-of-life gains are inconsistent. Newer techniques (photoselective vaporisation, Aquablation, Rezum, UroLift, prostatic artery embolisation) are not proven better than TURP for nocturia. **Counsel** that nocturia may respond sub-optimally, especially when it is the most bothersome symptom.

Overactive bladder

- Night symptoms are among the most refractory; daytime response does not predict night benefit. Antimuscarinics may reduce nocturnal urgency and raise night capacity. Tolterodine (night voids as primary endpoint) was negative overall but reduced urge-related night voids; **fesoterodine** (only trial with nocturnal urgency as primary endpoint) was positive. Patients voiding below capacity (**raised NBCi**) respond best to an evening dose.
- β3-agonists (mirabegron, vibegron): marginal reduction (≈0.1–0.2 voids vs placebo); none studied with nocturia as primary outcome.
- **Behavioural therapy** should precede drugs and matches or beats them for nocturia. Adding drugs adds little; group classes or video training cut cost.
- Third-line: onabotulinumtoxinA beats placebo by a fraction of a void; percutaneous tibial nerve stimulation beats sham and ≈ tolterodine; sacral neuromodulation data sparse. Nocturia never a primary outcome.
- Antimuscarinics are on the **Beers Criteria** list for ≥65 years — the core nocturia population.
- GSM (LUTS plus genital and sexual symptoms of oestrogen deficiency): nocturia via LUTD, night production and poor sleep; vaginal oestrogen may help, ospemifene potentially.

Therapy	EAU Male LUTS	EAU Female LUTS
α1-blocker	Offer for nocturia with LUTS (weak)	—

Therapy	EAU Male LUTS	EAU Female LUTS
5ARI	Offer with LUTS and prostate >**40 mL** (weak)	—
phosphodiesterase type 5 inhibitor (PDE5I)	**Do not offer**	—
Antimuscarinic	Offer with OAB (weak)	Offer after counselling (**strong**)
PFMT; vaginal oestrogen	—	Offer (**strong**); offer (weak)

> **KEY POINT** Lower urinary tract drugs act on capacity and outflow, but most nocturia is driven by production — so effects are statistically significant but clinically small. Treat the mechanism, not the organ.

10 · MEDICATIONS CAUSING NOCTURIA

Mechanism	Drugs
↑ Urine output	Diuretics; selective serotonin reuptake inhibitors (SSRIs; block antidiuretic hormone); **calcium channel blockers** (oedema, ↑ ANP); tetracycline; lithium (↓ aquaporin-2); **sodium-glucose cotransporter 2 (SGLT-2) inhibitors** (glycosuria)
Insomnia / central nervous system	Stimulants; α-blockers, β-blockers, methyldopa; salbutamol, theophylline, decongestants; corticosteroids, thyroxine; SSRIs, monoamine oxidase inhibitors; carbidopa; phenytoin
Bladder toxicity	Ketamine; tiaprofenic acid; cyclophosphamide

The three groups mirror the production, sleep and bladder arms. Note α-blockers both treat nocturia and may cause insomnia. Reconcile drugs and dosing times; first adjust timing or frequency, or substitute.

11 · MANAGEMENT ALGORITHM AND REFRACTORY NOCTURIA

ALGORITHM
History + examination + diary + basic tests → identify Sleep CALM factors → optimise them (multidisciplinary) → conservative measures → antidiuretic or timed diuretic in selected refractory patients — diary at every step

- **Multicomponent therapy**: causes interact, so treat them together (EAU **"Shared Care Pathway"**).
- **Target night volume even when capacity is small**: excess production best predicts frequency and response; small-capacity patients improved as NUV fell with capacity unchanged.

Desmopressin

- AVP normally rises in sleep; a **blunted rise** occurs in some NP patients.
- **Desmopressin** (1-deamino-8-D-arginine vasopressin, DDAVP): selective **V2 agonist** — antidiuretic without pressor effect. The **only US-approved nocturia drug**: **≥2 voids per night due to NP**. Level 1b evidence.

	Nasal spray	Orally disintegrating tablet (ODT)
Dose	1.66 mcg if <65, no risk; ≥65 or at risk → **0.83 mcg**	**Sex-specific**: women **25 mcg**; men **50 mcg**
Timing	≈30 min before bed	Sublingual ≈60 min before bed

- **Women are more sensitive**: the V2 receptor gene is on the **X chromosome**. In men 75 mcg added hyponatraemia, not efficacy.
- †Advise minimal fluid from 1 h before until 8 h after the dose. †Contraindicated with past or current hyponatraemia, polydipsia, loop diuretics or systemic/inhaled glucocorticoids, estimated

glomerular filtration rate <50 mL/min, syndrome of inappropriate antidiuretic hormone and heart failure.

- **Hyponatraemia** risk factors (sodium <**130 mmol/L**): **age ≥65**, baseline sodium <135, creatinine clearance <60 mL/min, doses above approved. Most occur within **3 weeks**.
- **Sodium monitoring**: EAU — baseline → **day 3** → day 7 → 1 month → periodically; US label — baseline → within 7 days → ≈30 days. EAU Female: avoid if baseline sodium low. A proposed protocol from pooled tablet data — baseline sodium in all, stop if <135 mmol/L, repeat at 7 and 30 days only if ≥65 or at risk — would have prevented every case below 130 at approved doses.
- **Low-dose rationale**: the median effective dose is far below that for maximal antidiuresis, so low doses still work and avoid antidiuresis outlasting sleep.
- EAU Male: desmopressin for NP <65; **low dose** >65 with benefit–risk discussion. **Age 65 is the pivot.**

Timed diuretics

- Furosemide acts for 6–8 h. Evening dosing **causes** NP; **morning or early-afternoon** dosing treats it (level 1b) by completing natriuresis before bed.
- Population links between diuretics and nocturia are mostly confounding by volume-overload disease. Small trials (bumetanide 4–6 h, furosemide 40 mg 6 h before bed) consistently cut voids. Monitor sodium.
- **Trap**: afternoon furosemide + bedtime desmopressin worked in a trial, but loop diuretics are a **contraindication** to desmopressin in US labelling.
- **Future**: phenotype night urine — predominant water diuresis → desmopressin; predominant natriuresis → afternoon diuretic.

> **KEY POINT** Refractory nocturia is treated by moving urine production out of the night. Either way the price is sodium balance, so dose, age and monitoring define safe use.

12 · LANDMARK TRIALS

Trial	Design	Result	Take-home
MTOPS	Doxazosin, finasteride, combination, placebo	Doxazosin and combination beat placebo; finasteride did not	α-blocker drives benefit
VA Cooperative	Terazosin, finasteride, combination, placebo	Only terazosin separated	5ARI alone does not help
CombAT	Tamsulosin, dutasteride, combination	Combination −0.5 vs −0.3/−0.4	Significant but small
Fesoterodine randomised controlled trial (RCT)	Nocturnal urgency primary endpoint	Positive	Helps urgency-driven night voids
Nasal spray phase III	0.83/1.66 mcg vs placebo	Both beat placebo	Basis of approval
ODT sex-specific RCTs	Women 25; men 50/75 mcg	Positive; 75 mcg more hyponatraemia	Sex-specific dosing

13 · MCQ ONE-LINERS

1. ICS 2019 nocturia: voids during the main sleep period, any cause.
2. First morning void: volume in NUV, counted as daytime frequency.
3. Mnemonic for causes: Sleep CALM.
4. NP (ICS): NPi >33% in older adults, or NUP >90 mL/h.
5. Global polyuria: >40 mL/kg/24 h.
6. Nocturia index: NUV / MVV.
7. NBCi = ANV − PNV; >2 severe.
8. Most important diary sleep measure: FUSP.
9. CPAP can reduce nocturia by ≈2 voids.
10. Only US drug approved for nocturia: desmopressin.
11. ODT doses: women 25 mcg, men 50 mcg.
12. EAU sodium checks: baseline, day 3, day 7, 1 month.
13. Loop diuretics contraindicate desmopressin.
14. Most first voids occur within 2–3 h of sleep onset.

5. Commonest cause of adult sleep disruption: nocturia.
6. Gold standard severity measure: diary-recorded ANV.
7. Krimpen study showed NPi discriminates poorly.
8. Nocturnal-only diary cannot diagnose global polyuria.
9. OSA raises night output via atrial natriuretic peptide.
10. STOP-BANG cut-offs: BMI 35, age 50, neck 40 cm.
11. Normal nocturnal blood pressure dip: 10–20%.
22. Calcium channel blockers cause nocturia via oedema.
23. PDE5Is are not offered for nocturia.
24. Isolated nocturia is not a LUTS.
25. 5ARI recommendation needs prostate >40 mL.
26. V2 receptor gene lies on the X chromosome.
27. Main hyponatraemia risk factor: age ≥65.
28. Diuretic timing for NP: morning or early afternoon.
29. Investigate small night capacity when NBCi >1.3.
30. Copeptin testing is superseding water deprivation for DI.

123 Surgical Procedures for Sphincteric Incontinence in the Male: Artificial Urinary Sphincter and Perineal Slings

Reading time ≈ 26 min (slow pace)

1 · PATHOPHYSIOLOGY AND DEVICE MECHANISMS

- Male **sphincteric (stress) urinary incontinence (SUI)** needs failure of **both** sphincters: internal (bladder neck, prostatic urethra) and external (rhabdosphincter). Radical prostatectomy (RP), the commonest cause, removes the internal sphincter by design, so continence rests entirely on the rhabdosphincter. Others: pelvic radiotherapy (RT), pelvic fracture urethral injury, neuropathy.
- Vesicourethral anastomotic stenosis (VUAS): its treatment can damage the membranous urethra and cause SUI.
- Work-up (cough-test grading, pad test, cystoscopy, urodynamics), conservative care and sling-versus-sphincter overview → *Voiding Dysfunction book, Ch 106*.

Device	Mechanism	Site	Adjustable	Needs residual sphincter
Artificial urinary sphincter (AUS, AMS 800)	Circumferential compression, patient-cycled	Bulbar urethra or bladder neck	No (pressure chosen at surgery)	No
Advance / Advance XP	**Repositioning** and lengthening, no compression	Bulbar, transobturator	No	**Yes**
Virtue (quadratic)	Prepubic compression + elevation	Bulbar, four arms	No	Less dependent
Adjustable transobturator male system (ATOMS)	Cushion compression	Bulbar, transobturator	Yes — scrotal port	Less dependent
ProACT (Adjustable Continence Therapy)	Two para-urethral balloons, lateral compression	Bladder neck	Yes — scrotal ports	—

- **AUS**: cuff (≈2 cm compression length), scrotal pump, pressure-regulating balloon (PRB) that sets cuff pressure. Squeezing the pump empties the cuff; it refills passively from the PRB †over a few minutes. Open cuff lowers resistance, so men without detrusor contraction can void by straining/Credé.
- Transobturator sling: raises leak point pressure **without obstruction** by elevating and elongating the bulb and rhabdosphincter.

> **KEY POINT** Every device either compresses the urethra or repositions a functioning sphincter. Trap: ATOMS and Virtue are "slings" but act by compression — only the Advance depends absolutely on a coapting residual sphincter.

2 · INDICATIONS AND DEVICE SELECTION

- Indication: **irreversible sphincter incompetence** with bothersome urinary incontinence (UI) after conservative care. Do not delay beyond **6 months** for severe/gravitational SUI, especially with a cystoscopic sphincter defect.
- Choice: severity, RT, prior urethroplasty, bladder function, cystoscopic coaptation, dexterity, cognition, life expectancy, preference.
- Severity (24-h pad weight): mild **<150 g**, moderate **<400 g**. AUS preferred if >250 g or after RT (MASTER trial → *Voiding Dysfunction book, Ch 106*).

Stratum (American Urological Association/Society of Urodynamics, Female Pelvic Medicine & Urogenital Reconstruction [AUA/SUFU])	Recommendation
Severe UI or **any RT**	AUS first-line; sling only with counselling
Moderate, no RT	AUS or male sling; adjustable balloons in specialised centres
Mild, no RT	Continue pelvic floor muscle training (PFMT), sling, possibly AUS

- Counsel all: expect ~1 thin pad/day; AUS lifespan **7–10 years**, replaceable; failed slings mostly go to AUS.
- **Sling requirements**: mild–moderate SUI, contractile bladder (hypocontractile → AUS, as it cannot overcome added resistance), well-vascularised non-irradiated urethra, coapting sphincter on cystoscopy.
- **AUS requirements**: dexterity and cognition to cycle the pump, healthy urethra, adequate compliance.

Absolute — need an alternative	Relative
Low-compliance bladder threatening kidneys; low capacity (frequency/urgency incontinence); urethral tissue too poor → clamp, catheter, bladder neck closure, diversion	Need for frequent transurethral work — bladder cancer, refractory VUAS, radiation cystitis (repeated instrumentation → erosion, infection)

- Radiation cystitis: hyperbaric oxygen and **6 months** free of bleeding before AUS.

> **KEY POINT** Each device's contraindication mirrors the other's requirement.

3 · THE HIGH-RISK ("FRAGILE") URETHRA

- Fragile urethra = raised cuff-erosion risk. Strong factors: **prior RT, prior erosion/infection, previous urethroplasty**; also hypogonadism, catheterisation, 3.5-cm cuff after RT, tandem distal cuff. Debated: coronary disease, age >80, diabetes, smoking.
- RT: microvascular spongiosal injury; erosion earlier and commoner.
- Prior erosion: 5-year repeat erosion approaches **50%**. Urethroplasty: worse than RT — transection divides bulbourethral arteries (loses dual blood supply).
- Countermeasures: optimise medically, **transcorporal (TC) cuff**, PRB **51–60 cm H_2O**, cuff 0.5 cm larger than measured, nightly deactivation.

> **KEY POINT** Erosion = spongiosal ischaemia under constant pressure. Every risk factor reduces perfusion; every countermeasure adds tissue (TC tunica) or lowers pressure — never squeeze harder.

4 · OPERATIVE TECHNIQUE

Preparation (all implants)

- General/spinal, low lithotomy, chlorhexidine–alcohol preparation. Urethral catheter optional (identifies urethra).
- Intravenous antibiotics within **60 min** of incision: aminoglycoside 5 mg/kg + first/second-generation cephalosporin or vancomycin. Sterile urine beforehand.

Bulbar cuff AUS (perineal)

- Site: **proximal bulbar urethra, proximal to corporal fusion** — largest spongiosal diameter, safe circumferential dissection, cuff not opened by sitting; distal sites kept for revision.

ALGORITHM
Midline perineal incision → Colles fascia → bulbospongiosus divided → incise Buck fascia where it reflects off the urethra onto the diverging crura → 2-cm tunnel dorsal to Buck fascia over the urethral roof under vision → right-angle clamp → measure circumference → cuff (usually 4.0–4.5 cm) → tubing through bulbospongiosus deep to Colles fascia

- Sizing: a consistent method matters more than catheter in/out; cuff slightly tighter than tape; less tight cuffs continent better. Sizes 3.5–11 cm. **3.5-cm cuff** (atrophy, very distal) raises erosion after RT → TC and/or 51–60 PRB.
- **PRB**: abdominal (lower-quadrant incision → split muscle → preperitoneal pocket → close fascia round tubing) or via external ring (perforate inguinal floor, or ectopic submuscular above transversalis fascia). Avoid the blind ring route after mesh hernia repair or cystectomy. Fill **23 mL** saline or isotonic diluted contrast (not if iodine allergy).
- **Pump**: anterior scrotum, subdartos. Quick connectors.
- Cycle and urethroscopy: closed cuff → slight blanching; open → minimal coaptation (excess → larger cuff).
- Leave **deactivated 4–6 weeks** — essential for healing without erosion. †Deactivate the cuff (open) before any later urethral catheterisation or endoscopy.

PRB (cm H_2O)	Use
51–60	High-risk bulbar (RT, prior erosion, atrophy) — trades continence for safety
61–70	Standard bulbar cuff
71–80	Bladder neck cuff

Variants

- **Tandem cuff** (incomplete occlusion; up to 11% leak with one cuff): new cuff ≥1.5–2 cm distal → join tubing with metal Y-connector → add 3 mL → deactivate 6 weeks. Not first-line — distal cuff erodes more, uses up revision sites.
- **Transcorporal cuff** (George Webster 2002 MCQ): parallel longitudinal tunical incisions on each corpus beside the spongiosum → dissect through intercorporal septum → ventral tunica stays on dorsal urethra → imbricate tunical edges → measure, place. For urethra <3.5 cm, revisions, prior erosion/urethroplasty (use a fresh site). Adds bulk to the thin dorsal spongiosum and avoids urethral injury; erections usually preserved; more transient retention; erosion benefit unproven.
- **Bladder neck AUS**: prostate intact (neuropathic, myelomeningocele); contraindicated after RP; open or robotic vesicoprostatic dissection; cuff ≥8 cm, PRB 71–80; less erosion and atrophy.

Male slings

ALGORITHM
Advance XP: perineal incision → open bulbospongiosus → mark and divide perineal body 1–2 cm (releases bulb) → stab incision one fingerbreadth below adductor longus tendon → helical trocar outside-in (two "pops") onto fingertip at spongiosum–ramus angle → mesh tension-free, 4 absorbable sutures to spongiosum → tension arms equally under cystoscopy until mucosa coapts (bulb moves 3–4 cm proximally) → bury arms, layered closure

- **ATOMS**: bulbospongiosus **left intact**; needles round ischiopubic ramus; cushion under bulb; subdartos scrotal port, 9 mL; adjusted later in clinic.
- **ProACT**: fluoroscopic perineal trocars to bladder neck; balloons; scrotal ports.
- Advance is tensioned once, at surgery; adjustable devices titrate in clinic at the cost of a port and more complications.

5 · AUS COMPLICATIONS AND MANAGEMENT

Retention

ALGORITHM
Confirm cuff deactivated → small urethral catheter 24–48 h → fails second trial → **suprapubic cystostomy** (ultrasound/fluoroscopy-guided, avoid PRB) → retention for weeks = undersized cuff → replace cuff

- Commoner with TC and 3.5-cm cuffs. Late retention → endoscopy + urodynamics.

Infection

- **1–3%** primary, up to 10% high-risk; *Staphylococcus epidermidis*, *S. aureus*. Antibiotic-coated (InhibiZone) devices have not reduced AUS infection.
- **Scrotal pain = key early sign** (± erythema, pus). Antibiotics fail → explant all; reimplant 3–6 months.
- Immediate salvage (non-eroded): remove all → antiseptic irrigation → new device; not with sepsis, erosion, immunosuppression or gross pus.

Urethral erosion

- **2–15%**; up to 35% with several risk factors. Remove **all components** (assume infected); loose closure/secondary intention. Post-erosion stricture 8–35%.

Urethral loss	Repair
<50%	Catheter (~3 weeks), urethrogram before removal
>50% / complete	Unresolved: in situ ventral approximation (fewer strictures than catheter alone) or mobilisation + anastomosis; pseudocapsule closure does not prevent stricture

- Reimplant at **3–6 months** after confirming patency, at a **new site**; TC ± low-pressure PRB; nightly deactivation. Repeat erosion up to 35%.

Urethral atrophy

- Chronic spongiosal compression → **gradual** recurrent leak; a commonest cause of revision; cystoscopy shows incomplete coaptation.
- Downsize in place → if already 4.0 cm, move proximally, or distally with TC; or tandem. **Never simply raise PRB pressure** — erosion.

Mechanical failure

- <3 years: replace failed part (pump or cuff). **>3 years: replace entire device** (PRB usual culprit; slow leaks hidden).

Recurrent leak after AUS	Clue → confirm
Deactivated	Sudden; reactivate (squeeze pump edges / press button)
Fluid leak	Sudden; PRB shrinks on film/ultrasound
Atrophy	Gradual; PRB volume normal (>20 mL) → cuff options
Erosion / detrusor overactivity	Cystoscopy / urodynamics

KEY POINT Tempo points to cause: sudden = device, gradual = urethra. Every complication is managed to keep a salvageable urethra for the next cuff.

6 · SLING COMPLICATIONS AND FAILURE

Device	Commonest	Erosion / explant
Transobturator	Transient groin pain, wound issues; retention/urgency 4–10%	0–0.4% — very rare

Device	Commonest	Erosion / explant
ATOMS	More complications	Erosion/infection ~10%; explant 10–16%
ProACT	**Perforation** (~5%); migration; revision 22%	~4%

- Cystoscopy in every worsening sling (mesh erosion). Infection/erosion → explant, AUS after 3–6 months.
- Failed sling → **AUS = gold standard**, results as for primary AUS; repeat Advance only if AUS refused (~56%).

7 · SPECIAL CIRCUMSTANCES

- Stricture at the cuff may herald erosion. VUAS proximal to cuff → balloon or holmium laser via flexible scope; large scopes → uncouple cuff.
- AUS + inflatable penile prosthesis (IPP): synchronous = metachronous; metachronous → AUS first.
- **Climacturia**: sling or AUS; with erectile dysfunction → **mini-jupette** graft at IPP.

8 · OUTCOMES

Device	Outcome	Weak point
AUS	Social continence (0–1 pad) 55–77%; satisfaction ~90%	Revision 10–29% at 3 years
Advance XP	Cure ~60%, durable	RT, severe UI, prior surgery
ATOMS / ProACT	~60–67% dry	Complications, limited data

- Cure clusters at 55–65%; devices differ in durability, erosion and eligibility, not cure.

9 · MCQ ONE-LINERS

1. AUS = gold standard for all SUI severities.
2. Only Advance sling depends absolutely on residual sphincter.
3. Any prior RT → AUS first-line.
4. Detrusor hypocontractility → AUS, not sling.
5. Standard bulbar PRB: 61–70 cm H_2O.
6. Bladder neck PRB: 71–80 cm H_2O.
7. AUS activation delayed 4–6 weeks.
8. Transcorporal cuff introduced by Webster, 2002.
9. Bladder neck AUS contraindicated after RP.
0. Strong erosion risks: RT, prior erosion, urethroplasty.
11. Repeat erosion after prior erosion approaches 50% at 5 years.
12. Early AUS infection sign: scrotal pain.
13. Atrophy: downsize cuff, never raise PRB pressure.
14. Mechanical failure >3 years: replace whole device.
15. Retention after AUS: suprapubic cystostomy after second failed trial.
16. Commonest ProACT complication: perforation.
17. PRB fill volume: 23 mL.
18. Mild SUI: <150 g/24 h.
19. Erosion: explant all components.
20. Failed sling → AUS.

★ Putting it together: nocturia and male stress incontinence

Reading time ≈ 3 min (slow pace) · Integrates Ch 112, 123 · use these as the skeleton of long-essay answers

A · NOCTURIA: THREE QUESTIONS FROM THE BLADDER DIARY

ALGORITHM

1. Global polyuria? 24-h output >40 mL/kg → look for diabetes mellitus or insipidus, polydipsia — always investigate.

2. Nocturnal polyuria? Night fraction >20% (young) or >33% (older adults) → heart failure, sleep apnoea, oedema states, evening fluids; consider desmopressin with sodium monitoring.

3. Reduced night capacity? Nocturnal bladder capacity index >1.3 → overactive bladder, obstruction, incomplete emptying; treat the lower urinary tract. Screen sleep disorders in all ("Sleep CALM") (→ *Ch 112*).

B · MALE STRESS INCONTINENCE AFTER PROSTATECTOMY

KEY POINT Pelvic floor training first, but do not delay surgery beyond about 6 months for severe leakage. Grade by 24-h pad weight (mild <150 g, moderate <400 g). Mild–moderate leakage, contractile detrusor, no radiotherapy → male sling; heavy leakage (>250 g), any radiotherapy or a failed sling → artificial urinary sphincter (cuff, pressure-regulating balloon, scrotal pump). Erosion follows spongiosal ischaemia — risk rises after radiotherapy, previous erosion or urethroplasty; counter it by adding tissue (transcorporal cuff) or lowering pressure, never a tighter cuff. † An eroded or infected device is removed (→ *Ch 123; Voiding Dysfunction book, Ch 106*).